Q1:

```
`default_nettype none  //ensure undeclared variable aren't inferred for you.

module endian(
   input wire [7:0] data,             //  data to be sent
   input wire big_endian,             //  programmable switch, if true, outputs be in big endian format
   output logic [7:0] send_data       //  send_data[0] will be first bit sent
   );
   always_comb begin
      if(big_endian) begin
        assign send_data = {data[7], data[6], data[5], data[4], data[3], data[2], data[1], data[0]};
      end else begin
         assign send_data = data;
      end
    end
endmodule

`default_nettype wire
```

Q2:

```
`default_nettype none  //ensure undeclared variable aren't inferred for you.

module total_system(
   input wire [2:0] a_in,
   input wire [5:0] b_in,
   output logic [4:0] f_out
   );
   logic [2:0] s_out_0;
   logic [1:0] s_out_1;
   logic [1:0] s_out_2;
   logic [3:0] s_out_3;
   logic [1:0] s_out_4;
   logic [4:0] s_out_5;
   m0 module_0 (.s_in(a_in), .s_out(s_out_0));
   m1 module_1 (.a_in(s_out_0), .b_in(b_in), .s_out(s_out_1));
   m2 module_2 (.a_in(s_out_0), .b_in(b_in), .s_out(s_out_2));
   m3 module_3 (.a_in(a_in), .b_in(b_in), .s_out(s_out_3));
   m4 module_4 (.a_in(s_out_1), .b_in(s_out_2), .s_out(s_out_4));
   m5 module_5 (.a_in(s_out_4), .b_in(s_out_3), .s_out(s_out_5));
   assign f_out = s_out_5;
  //  YOUR CODE HERE
```

```
endmodule

`default_nettype wire
```